A Hybrid Perching Mechanism for Aerial Robots: Branch and Surface Perching Robust to Large Surface Orientation Misalignment

Sudheera Akalanka Kariyawasam[1], Mahmud Hasan Saikot[1], Bo Cheng[2], and Jianguo Zhao[1]

***Abstract*——Perching has recently emerged as a key capability in aerial robotics, enabling substantial gains in energy efficiency and operational endurance for long-duration missions. In this work, we present a multifunctional perching mechanism capable of both branch and surface perching. For branch perching, the mechanism uses four simultaneously-actuated claws to grasp branches with diameters up to 35 mm. For surface perching, it leverages a bistable mechanism with five suction cups, allowing perching onto flat smooth surfaces with large orientation misalignment (with surface tilt angles of up to 80°). The mechanism is fabricated using a combination of multimaterial 3D printing and silicone molding, resulting in a lightweight prototype with a mass of 32 g. Experimental results show that the proposed design achieves 6.4× adhesion success compared to a bistable mechanism with a single suction cup. Furthermore, real-world perching experiments onto cars and tree branches demonstrate the effectiveness of the mechanism in different environments, highlighting its potential for energy-efficient and adaptable aerial robotic operations.**

***Index Terms*—Aerial Systems: Mechanics and Control; Aerial Systems: Applications; Mechanism Design**

I. INTRODUCTION

RECENT advances in aerial robots have enabled transformative applications in agriculture, infrastructure inspection, and defense. Despite tremendous progress, limited endurance remains a key bottleneck. Since many missions do not require the drone to stay airborne at all times [1], perching can allow aerial robots to attach to the environment and deactivate propulsion, extending operational time while minimizing acoustic impact.

Perching has therefore emerged as an active research area. Existing strategies can be generally classified by substrate type: (i) surface perching, which involves landing on flat, inclined, or curved structures (e.g., windows, ceilings, hulls), and (ii) branch perching, which focuses on attachment to rod-like structures (e.g., tree branches, pipes) [2]. Here, we briefly highlight some recent representative approaches; see [2] for a comprehensive review.

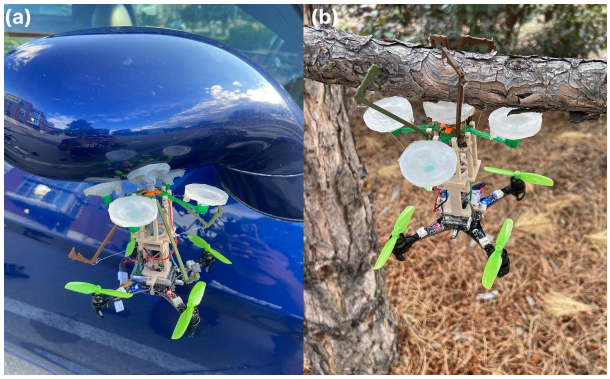


Fig. 1. Hybrid perching demonstrations. (a) Surface perching to a car side-view mirror. (b) Branch perching onto a tree branch.

Surface perching mechanisms include suction cups, electrostatic adhesion, polymer adhesives, microspines, and magnetic adhesion [3], [4]. Suction cups and electroadhesion work well on smooth surfaces. Suction cups can withstand large forces but are not effective on rough surfaces [5]. Electroadhesion is not fully passive and may consume power after perching [6]. Dry or wet adhesives can work on non-smooth surfaces, but dry adhesives often degrade over time and are load-sensitive, whereas wet adhesives can adhere to a wider range of surfaces [7], [8]. Microspines can anchor reliably by penetrating the surface, and magnetic adhesion only works on metallic surfaces. However, both are limited by surface texture and material compatibility [9], [10].

Branch perching mechanisms typically employ grippers, tethers, claws, or spines [11]–[13]. Representative designs include passive microspine grapples that encircle cylindrical objects and sustain large loads [11]. Origami-based spines mounted on protective cages can engage irregular branches through passive reorientation [14]. Bird-inspired claw-and-tail designs enable vertical trunk perching by combining a front gripping module with a rear stabilizing support [12].

Despite significant progress, existing perching mechanisms generally face two key limitations. First, most are designed for a single substrate type, while real environments often demand hybrid capability (surface and branch perching). Second, it remains challenging to realize robust surface perching under large orientation misalignment, especially on tilted surfaces. In many scenarios, these surfaces may also be moving (e.g.,

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TABLE I
COMPARISON OF SOME TYPICAL PERCHING MECHANISMS

Mechanism	Weight	Surface perching	Misalignment handling	Branch perching	Actuation
Electrostatic [18]	2 g	Yes	–	No	Electrostatic
Adhesive pad [19]	–	Yes	Active	No	Passive
Bird claws [20]	250 g	No	N/A	Yes	Servo
Compliant cup [21]	–	Yes	Active (∼90°)	No	Vacuum
Suction cup [22]	315 g	Yes	Passive (limited)	No	Servo
Magnetic [23]	250 g	Yes	Active (tilt-rotor)	No	Servo
Avian-inspired [24]	478 g	No	N/A	Yes	Passive
Rolling graspers [25]	–	No	N/A	Yes	Servo
Gripper-based [26]	–	No	N/A	Yes	Servo
Tunable [15]	–	Yes	Active (∼35°)	No	SMA
Hybrid (This work)	**32 g**	**Yes**	**Passive (∼80°)**	**Yes**	**DC motor**

Notes: N/A = Not Applicable; – = Not Reported.

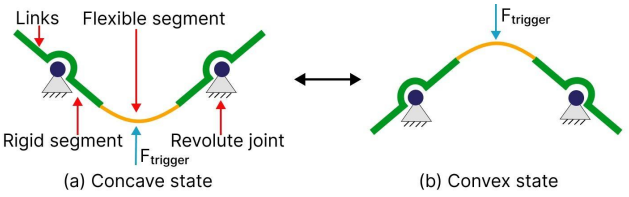


Fig. 2. Schematic illustration of the bistable mechanism with two stable states: (a) Concave state with the flexible segment positioned at the bottom. (b) Convex state with the flexible segment positioned at the top. The mechanism switches between these states when an external force is applied: F_{trigger} drives concave → convex, and F_{trigger} drives convex → concave.

ship decks at sea or other aerial robots in flight), making it extremely difficult to rely on precise control algorithms, as surface orientation can change rapidly. To address this challenge, mechanisms with mechanical intelligence have been investigated as they can passively accommodate misalignment. However, most existing mechanisms tolerate only small misalignments in one dimension, and larger misalignments still rely on active components [15] or sophisticated control [16], [17]. It remains largely unexplored whether a perching mechanism can handle large, three-dimensional misalignments.

To address these limitations, we present a hybrid perching mechanism capable of reliable perching on tree branches and smooth surfaces (Fig. 1). *The main contribution of this work* is a hybrid mechanism that integrates a motor-actuated four-claw gripper for branch perching with a bistable penta-suction-cup mechanism for smooth surface perching. Further, the surface-perching mechanism can passively attach to surfaces tilted up to 80° regardless of robot orientation without complex motion control. The mechanism is validated through experiments on smooth surfaces and branches with diameters up to 35 mm. We compare our mechanism with some existing ones in terms of functionality and performance in Table I.

The rest of this paper is organized as follows. Section II describes the design concept and working principle. Section III analyzes key design parameters. Section IV details the fabrication process. Section V presents the experimental results, and Section VI concludes the paper.

II. DESIGN AND WORKING PRINCIPLE

Our mechanism consists of two primary subsystems for surface and branch perching, respectively. Depending on the task, either subsystem can be engaged. The mechanism is mounted on top of the drone to keep the center of gravity at the drone's center. The following describes the design and working principle of each subsystem.

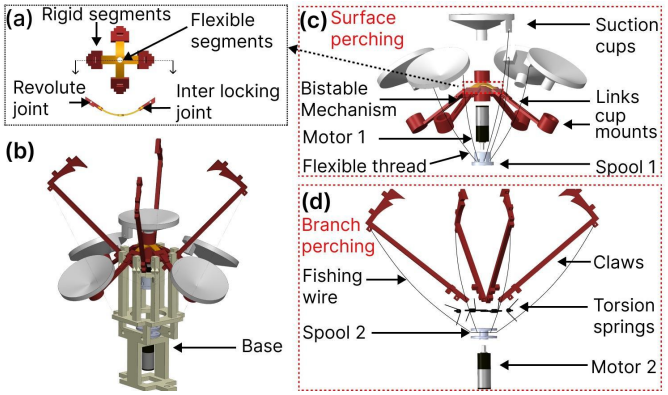


Fig. 3. The detailed design for the hybrid perching mechanism. (a) Top and section view of the spatial BM. (b) 3D model of the assembled mechanism. (c) Exploded view of the surface perching mechanism. (d) Exploded view of the branch perching mechanism.

A. Surface perching

For surface perching, we target passive attachment to conserve energy and avoid complex designs. This is realized by a bistable mechanism (BM) with two stable states, enabling fast attachment upon surface contact. The surface perching subsystem consists of five main elements: a spatial BM, a lightweight base, four rigid links, five suction cups, and a suction cup release mechanism (Fig. 3).

The spatial BM is constructed by integrating two planar BMs orthogonally and sharing a common section. To clarify its operation, the behavior of a planar BM is first described. A planar BM has two stable states: a concave and a convex state (Fig. 2). It consists of two rigid segments connected by a flexible segment, with the rigid segments mounted to a fixed base via revolute joints. In the concave state, applying an upward force F_{trigger} to the flexible segment transitions the mechanism to its convex state. Conversely, in the convex state, applying a downward force F_{trigger} reverts it back to the concave state (see Section III for trigger force analysis).

Combining two planar BMs orthogonally forms a spatial BM with two flexible segments and four rigid links (Fig. 3a). This spatial BM hosts five suction cups: one at the flexible-segment's center and one at each rigid-link's end. By generating a vacuum at the contact interface, the suction cups provide strong adhesion to smooth surfaces (Fig. 3c).

A release mechanism is designed to detach the suction cups from a surface. It uses motor-driven threads connected to each suction cup edge. Pulling the cup edge requires minimal force, as this area experiences the lowest vacuum pressure. The elastic threads do not deform the suction surface when slack. Upon motor actuation, the threads stretch and pull the suction cup edges, enabling effective release.

Based on the bistable penta-cup design, we illustrate how the mechanism enables perching onto smooth surfaces under

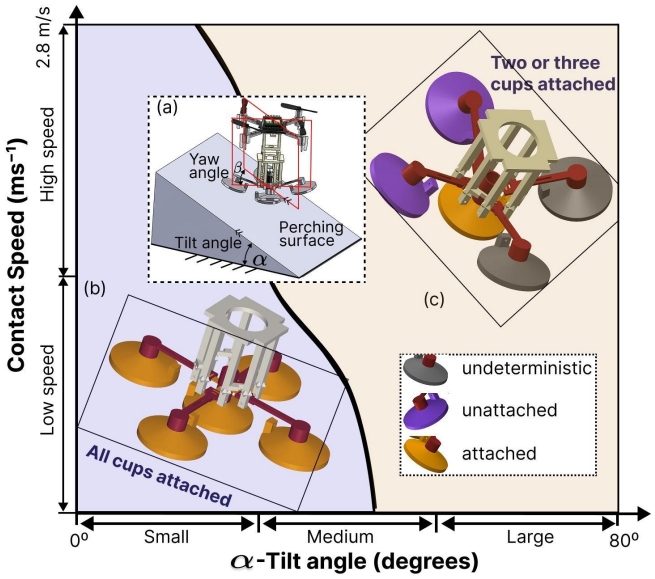


Fig. 4. Different scenarios for surface perching: (a) Schematic to show yaw and tilt angles. (b) Five-cup attachments with triggering. (c) Two- or three-cup attachments without triggering.

different conditions (Fig. 4a): (i) the surface tilt angle α, defined as the inclination of the perching surface relative to the ground; (ii) the drone's yaw angle β; (iii) the contact speed v; and (iv) whether the BM switches state. These conditions determine how many suction cups attach to the surface after perching. We broadly classify them into two categories based on α and v (Fig. 4). Due to design symmetry, roll angle is not considered. In all cases, the BM starts from the concave state when perching onto a surface below the mechanism.

In the first category, all five cups attach. This happens at small α regardless of v, or at medium α with low v. In this category, attachment is yaw-independent, and all five cups engage regardless of β. The center cup typically initiates contact, generating an impact force that triggers the BM state change, allowing the four side cups to engage and form full attachment (Fig. 4b).

In the second category, two or three cups attach without triggering the BM. In this category, the attachment process generally can be described as follows: the center cup initially establishes contact and serves as a pivot, inducing a rotational motion about the impact point that allows the engagement of the adjacent side cups, resulting in attachment without triggering the BM (Fig. 4c). The yaw angle determines how many cups will attach to the surface. Large yaw angles β ($\sim 45°$) lead to three-cup attachments, typically in combination with medium tilt angles α at high speeds or with large α. Small yaw angles β ($\sim 0°$) generate two-cup attachments under similar tilt conditions.

B. Branch perching

The branch perching subsystem comprises four rigid claws mounted to the base through spring-loaded revolute joints so that they are normally closed (Fig. 3d). A DC motor opens all claws simultaneously by winding cables attached to each

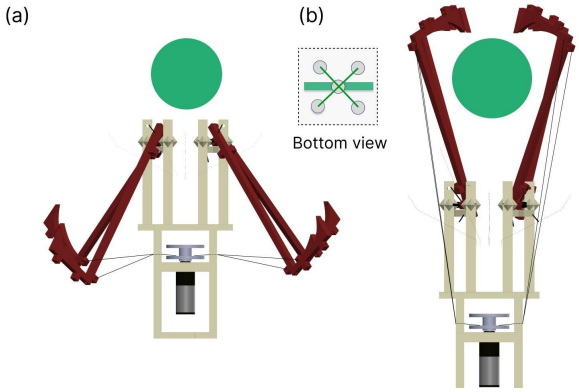


Fig. 5. Working principle for branch perching. (a) The mechanism approaches a branch with its claws open under motor actuation. (b) Perching onto the branch by enclosing the branch with claws under the torsional spring force.

claw onto a spool. When the spool unwinds, the mechanism can close due to the torsional springs, allowing the robot to perch onto structures like tree branches or grasp an object.

To enhance perching, a pair of teeth is integrated at each claw's tip. These teeth increase the frictional interaction with the branch surface. The geometry of the teeth is selected to ensure a secure and stable perch. In operation, the claws remain fully open until the branch is roughly centered between them. The claw opening angle adapts to the branch diameter to ensure effective engagement (Fig. 5a). Once aligned, the claws close simultaneously, establishing a secure grip on the branch surface (Fig. 5b). After closure, the gripper holds position without motor power due to spring-loaded claws.

III. ANALYSIS OF TRIGGERING FORCE

This section presents the design process of the perching mechanism, focusing on the analysis of the BM and the selection of its design parameters (Fig. 6a). The triggering force of the spatial BM is a critical parameter for surface perching: too large a value requires high contact velocity to trigger the mechanism, while too small a value risks auto-triggering under sudden drone acceleration. We investigate the triggering force for a planar BM, as the spatial BM is an extended configuration of the same principle.

The triggering force F_{trigger} depends on several parameters, but for simplicity, we fix the length between the two revolute joints L and the rigid link length d (Fig. 6b), leaving the flexible segment length a as the design variable. To evaluate how F_{trigger} varies with a, we use the energy approach by considering the difference of stored strain energy in the flexible beam between the initial state (U_1, left of Fig. 6b) and the maximum energy (U_2, right of Fig. 6b) when the transition happens. The energy difference $U_2 - U_1$ can be approximated with $F_{\text{trigger}} D/2$ where D is the displacement of the flexible segment's center point from the initial state to the unstable state. Therefore, we can obtain F_{trigger} using

$$F_{\text{trigger}} \approx \frac{2(U_2 - U_1)}{D} \tag{1}$$

Based on the geometric relationship, we have $D = d \sin\theta + H$ where θ and H are labeled in Fig. 6b. We approximate the

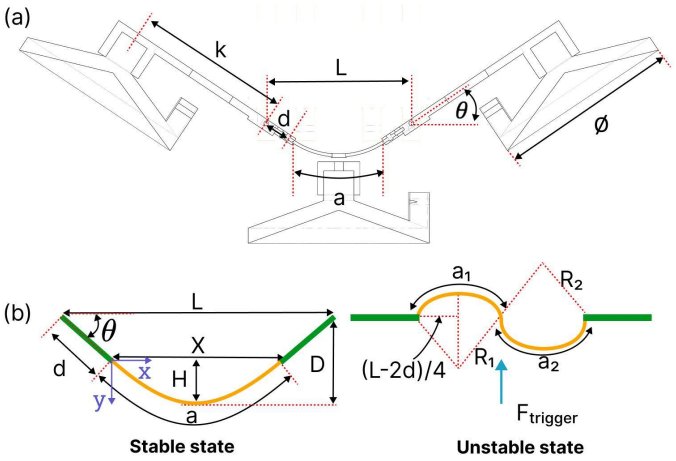


Fig. 6. The design parameter analysis of surface perching mechanism. (a) Design parameters of the mechanism. (b) Parameters of the flexible segment for stable and unstable states.

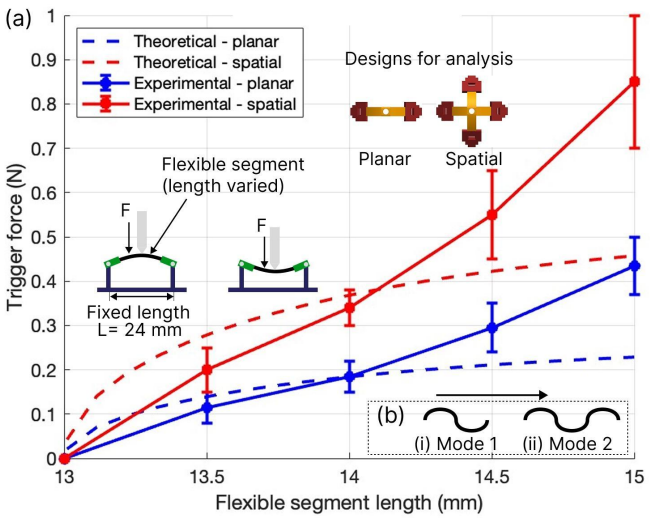


Fig. 7. Trigger force analysis. (a) Theoretical evaluation and experimental validation of triggering force for both planar and spatial BM. (b) Mode shapes for unstable states for varied segment lengths (i) mode 1: sine curve; (ii) mode 2: 1.5x sine curve.

shape of the flexible segment as a parabolic curve in vertex form $y = -(x - X/2)^2 tan\theta/X + X tan\theta/4$, where $X = L - 2d\cos\theta$ is the horizontal span, leading to $H = 0.25X\tan\theta$.

The next step is to establish the relationship between θ and a, which can be obtained from the initial state by integrating the gradient of the parabolic function for a span length X.

$$X = \frac{2a\tan\theta}{\tan\theta\sqrt{1+\tan^2\theta} + \text{asinh}(\tan\theta)} \tag{2}$$

We can also obtain U_1 and U_2 as a function of a and θ [27]. For U_1 in the stable state, the flexible segment can be treated as a bending beam, and it can be expressed as $U_1 = 2Ba\sin^2\theta/X^2$, where B is the bending stiffness of the flexible segment. For U_2 in the critical unstable state, it can be derived from $U_2 = \frac{B}{2}(\frac{a_1}{R_1^2} + \frac{a_2}{R_2^2})$, where a_1, a_2 are unstable section lengths such that $a_1 + a_2 \approx a$ and R_1, R_2 represent the bending radii in the unstable state. To quantify energy analytically, we assume an ideal scenario where $R_1 = R_2 = R$ with $a_1 = a_2 = a/2$. Under such assumptions, R can be obtained by analyzing the geometric relationship of the stable state $R\sin(\frac{a}{4R}) = \frac{L-2d}{4}$ (Fig. 6b).

The above analysis is conducted considering only the planar BM. Since the spatial BM combines two planar BMs, its triggering force can be approximated as $2F_{trigger}$.

Finally, the resultant triggering force $F_{\text{resultant}}$ needs to consider additional parts in the perching mechanism, including the suction cups and DC motor. $F_{\text{resultant}}$ varies according to how the mechanism is attached to the drone. If the mechanism is attached to the top, then $F_{\text{resultant}} = 2F_{\text{trigger}} + F_{\text{weight}}$, where F_{weight} is the force generated by additional parts, as $F_{\text{resultant}}$ needs to overcome F_{weight}. If the mechanism is attached to the bottom, then $F_{\text{resultant}} = 2F_{\text{trigger}} - F_{\text{weight}}$. Note that $F_{\text{weight}} = [(\frac{8k}{L} - 1)M - m]g$, where k is the link length (Fig. 6a), M and m are the mass for the suction cups and DC motor, respectively.

To validate the developed model, we first measure the bending stiffness of the flexible segment using a three-point bending test. The segment is fabricated from Thermoplastic Polyurethane (TPU) (width 3.5 mm, thickness 0.8 mm) to obtain a good quality 3D print (15% infill, Prusa XL). The resulting bending stiffness is measured to be 1.2×10^{-6} N m^2.

We validate the model by fixing $L = 24$ mm, $d = 5.5$ mm, varying a from 13 to 15 mm (step size 0.5 mm) for both planar and spatial BMs, and obtaining the triggering force for each scenario. For each value of a, we measure the triggering force using a force gauge (M5-2, Mark-10) five times. For both setups (planar and spatial), the triggering force increases with respect to a (Fig. 7a), with some discrepancy between the theoretical and experimental results. For the planar BM, the experimental force increases from 0.11 ± 0.03 N at a = 13.5 to 0.42 ± 0.04 N at a = 15. For the spatial BM, the corresponding values are 0.20 ± 0.03 N and 0.85 ± 0.10 N. These results indicate that both the mean values and the standard deviations increase with a. The discrepancy between theoretical and experimental values may be due to the unmodeled friction in the hinges and the idealized shape during the unstable state (Fig. 6b). The discrepancy grows with respect to a due to changes in the unstable mode shape (Fig. 7b). With a small a, mode 1 dominates the deformation, which is assumed in our model. As a increases, mode 2 emerges, which is not considered in our model, causing the observed deviation between theory and experiment [28], [29]. Since we target small a values with low triggering forces, the simplified model remains sufficiently accurate.

To minimize the mass of the mechanism, miniature silicone cups with a diameter of $\phi = 30$ mm are fabricated, leading to a mass of $M = 2.5$ g. For mounting, a link length of $k = 28$ mm is identified as optimal, as it minimizes the space between the center and edge suction cups while maintaining a lightweight design.

To minimize contact force for perching, $F_{\text{resultant}}$ should be as small as possible for both the bottom-mounted and top-mounted configurations. To prevent auto-triggering, $F_{\text{resultant}}$ should be close to zero in the bottom-mounted configuration. We select $F_{\text{resultant}}$ = 0.1 N to provide some safety margin. This

choice corresponds to $F_{\text{resultant}} = 0.48$ N when top-mounted. For this $F_{\text{resultant}}$, $F_{\text{trigger}} = 0.15$ N, corresponding to $a = 13.5$ mm.

For branch perching, the claw length determines the maximum branch diameter the gripper can accommodate. To maintain a lightweight and compact design, the claw length is constrained by the base height to avoid propeller contact when retracted. These dimensional constraints result in a claw length of 60 mm.

IV. FABRICATION

In this section, we detail the fabrication process of the perching mechanism, which includes the spatial BM fabricated using multimaterial 3D printing and the soft suction cups made from silicone molding.

We use a multimaterial 3D printer (Prusa XL) to print the spatial BM as a single piece. This piece includes rigid links, suction cup mounts, and rigid and flexible segments. The flexible segment is printed in TPU, while the remaining parts use Polylactic Acid (PLA). An interlocking joint provides a strong connection between the flexible and rigid segments (Fig. 3a). The base, claws, and motor spools are also 3D-printed separately in PLA (X1 series, Bambu Lab). The multimaterial piece is connected to the base via four pins serving as revolute joints.

The miniature suction cups are fabricated using a three-part mold: bottom, top, and wall molds. The bottom mold is coated with XTC-3D (Smooth-On) to achieve a smooth surface and a reliable vacuum seal. The wall mold is secured to the bottom mold to ensure proper sealing, followed by the top mold. Next, a properly mixed and degassed silicone (Ecoflex 00-50, Smooth-On) is poured into the mold, with vacuuming used to eliminate any trapped air bubbles. Curing is accelerated by placing the mold in an oven set to 60°C for 1 hour. After curing, the suction cup is demolded, and excess material is trimmed. Finally, these suction cups are attached to their mounts using silicone glue (Sil-Poxy, Smooth-On).

The spatial BM is assembled with motor 1 (NFP-612-136-0675, Precision Mini Drives) and spool 1, and the center cup is mounted (Fig. 3c). Flexible threads from the cup ends route to connect spool 1 on motor 1, and the assembled setup connects to the base via pins. Each claw is mounted to the base with torsion springs. Finally, motor 2 (NFP-612-136-0675, Precision Mini Drives) and spool 2 are integrated, connecting claws using fishing line to enable proper opening and closing (Fig. 3d).

V. EXPERIMENTS

We experimentally evaluate our hybrid perching mechanism for both surface and branch perching. For surface perching, we compared our mechanism (penta-cup design) with two other designs (Fig. 8). Both designs have a BM to ensure a fair comparison. One design (Fig. 8a) is a planar BM with a single suction cup placed at the center, as a single cup is commonly used in the literature. The second design (Fig. 8b) has three cups on a planar BM, one at center and one at each end, serving as a simplified planar case for the penta-cup design.

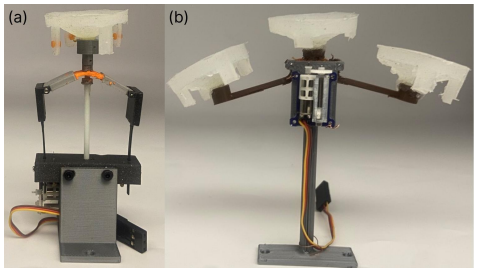


Fig. 8. Mechanisms used to compare perching performance. (a) Bistable single cup. (b) Bistable tri-cup.

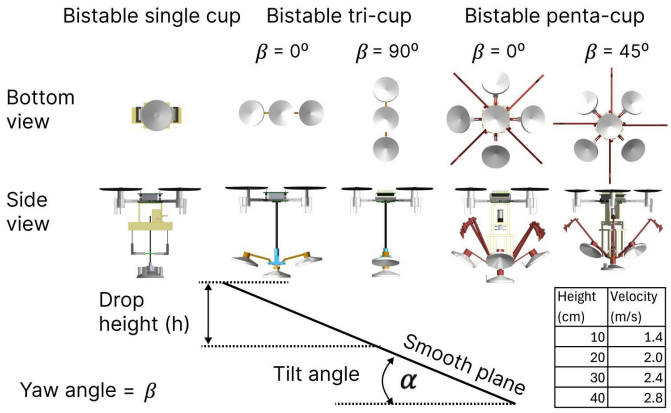


Fig. 9. Experimental setup for perching on smooth flat surfaces at different tilt (α) and yaw (β) angles with respect to different drop heights (h).

A. Perching onto flat surfaces with different orientations

We first evaluate all three designs to perch on tilted flat surfaces (a smooth glass sheet) by placing the perching mechanism beneath a quadrotor (Crazyflie 2.1 brushless, Bitcraze). Without loss of generality, we drop the quadrotor from different heights to maintain uniform conditions across all experiments to ensure a fair comparison.

We vary the surface tilt angle α by adjusting the glass sheet's inclination relative to the ground. We compare the performance of the tri-cup and penta-cup designs for two yaw angles β: 0° and 90° for the tri-cup, and 0° and 45° for the penta-cup, representing the endpoints of variation. For each β, five drop trials are performed at four heights (10, 20, 30, and 40 cm) to achieve different contact velocities (Fig. 9). Since the penta-cup weighs 32 g, additional weights are added to the other two mechanisms for a fair comparison.

To evaluate the perching performance, we define the strength level (SL) as

$$SL = \frac{1}{N}\sum_{i=1}^{N} P_i \tag{3}$$

where $N = 5$ is the number of tests, and P_i is the number of attached cups for the $i-$th test.

The performance of the *bistable single-cup* design is examined by varying α from 0° to 40°. The SL remained consistently at 1 across all drop heights for tilt angles up to 20°. Beyond this threshold, the SL gradually declines, reaching substantially lower values at 40° (Fig. 10). This trend

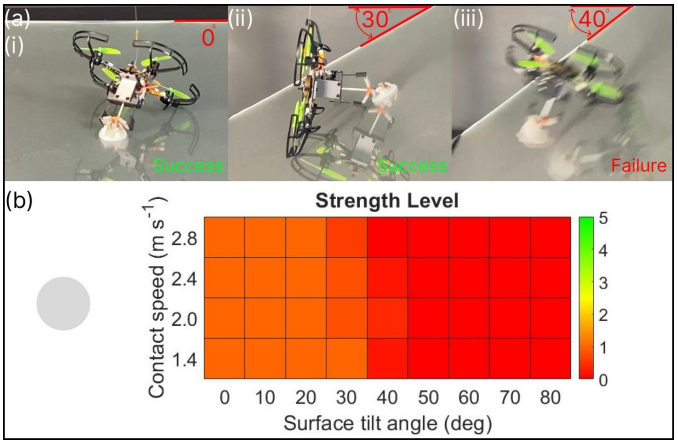


Fig. 10. Experimental results for the single cup mechanism. (a) Example perching at different tilt angles: (i) success at $\alpha = 0°$; (ii) success at $\alpha = 30°$; (iii) failure at $\alpha = 40°$. (b) Strength level vs tilt angle.

can be attributed to the triggering behavior of the BM: for α up to $20°$, the mechanism reliably activates the bistable transition, ensuring secure attachment. However, for α from $20°$ to $40°$, the mechanism attaches without triggering. Note that experiments for $\alpha > 40°$ are not conducted since the perching will fail (i.e., $SL = 0$), but we include the results for comparison with the other two designs. Furthermore, due to the circular geometry of the cup, the performance of this mechanism is unaffected by its yaw angle.

The *bistable tri-cup* mechanism is tested at $\beta = 0°$ and $\beta = 90°$. When $\beta = 0°$, the mechanism achieves SL values between 2 and 3 for tilt angles up to $40°$ (Fig. 11b). Higher contact speeds prevent the BM from triggering, resulting in attachment by only two cups. When $50° \leq \alpha \leq 80°$, the BM does not trigger, yielding reduced SL between 0.6 and 1.2. In this range, higher speeds (drop heights of 30–40 cm) further reduce attachment success, whereas at lower speeds the mechanism can attach with two cups (Fig. 11b). When $\beta = 90°$, the mechanism behaved similarly to when $\beta = 0°$ for tilt angles up to $40°$, achieving SL between 2 and 3. Beyond this range, SL decreases sharply, with negligible success at $50°$ (Fig. 11c). This is due to the absence of side suction cups in the $\beta = 90°$ direction, which reduces impact absorption and attachment capability at large tilt angles.

The *bistable penta-cup* mechanism is tested at $\beta = 0°$ and $\beta = 45°$. When $\beta = 0°$, the mechanism achieved SL between 3 and 5 for tilt angles up to $40°$. At higher contact speeds, however, the BM fails to trigger, resulting in attachment by only two cups. When $50° \leq \alpha \leq 80°$, the SL ranged between 1 and 3, with lower success rates at higher speeds (Fig. 12b). When $\beta = 45°$, the mechanism exhibits similar performance up to $\alpha = 40°$. For tilt angles up to $80°$, SL between 1 and 2.5 is achieved. In these cases, even without BM triggering, the system achieves stable attachment using three cups. At very steep angles (e.g., $\alpha = 60°$) and low drop heights (10 cm), the BM occasionally triggers, allowing attachment with all five cups (Fig. 12c). When compared to the tri-cup mechanism with $\beta = 90°$, this mechanism shows significantly improved performance in handling larger tilt angles.

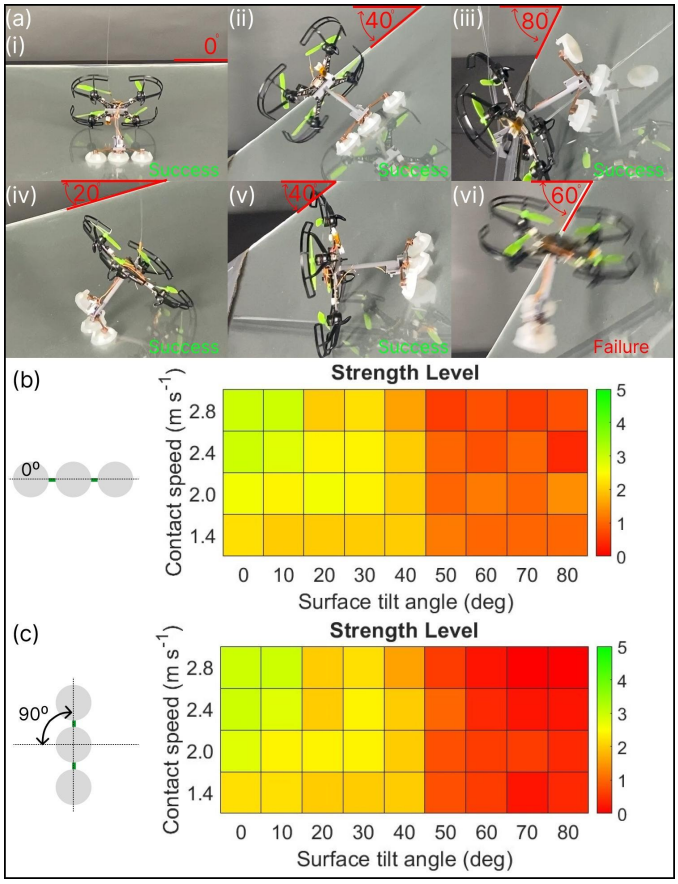


Fig. 11. Experimental results for the tri-cup mechanism. (a) Example perching at different tilt α and yaw angles β: (i) success at $\alpha = 0°$, $\beta = 0°$; (ii) success at $\alpha = 40°$, $\beta = 0°$; (iii) success at $\alpha = 80°$, $\beta = 0°$; (iv) success at $\alpha = 20°$, $\beta = 90°$; (v) success at $\alpha = 40°$, $\beta = 90°$; (vi) failure at $\alpha = 60°$, $\beta = 90°$. (b) Strength level variation for yaw angle of $\beta = 0°$. (c) Strength level variation for yaw angle of $\beta = 90°$.

To validate the drop test results, we conduct flight experiments with $\beta = 0°$ and $\beta = 45°$. However, to ensure stable flight using the flow deck beneath the Crazyflie, we mount the mechanism on top of the drone and accelerate it toward the surface for contact. The mechanism shows comparable in-flight performance, consistent with the drop test results (Fig. 13).

B. Release Testing

The developed detachment mechanism is also tested to demonstrate its functionality. Upon actuation of the motor, the flexible thread is tensioned, causing the suction cups to detach sequentially until the mechanism is fully released from the surface. When the motor rotates in another direction, the thread slackens, allowing the suction cups to return to their normal state, i.e., ready to fly and perch again. The mechanism requires approximately 5 s to detach from the surface and 5 s to return to its normal state (Fig. 14).

C. Perching onto branches

For branch perching, the holding force is determined by the claw-end geometry. Therefore, we evaluate the holding force

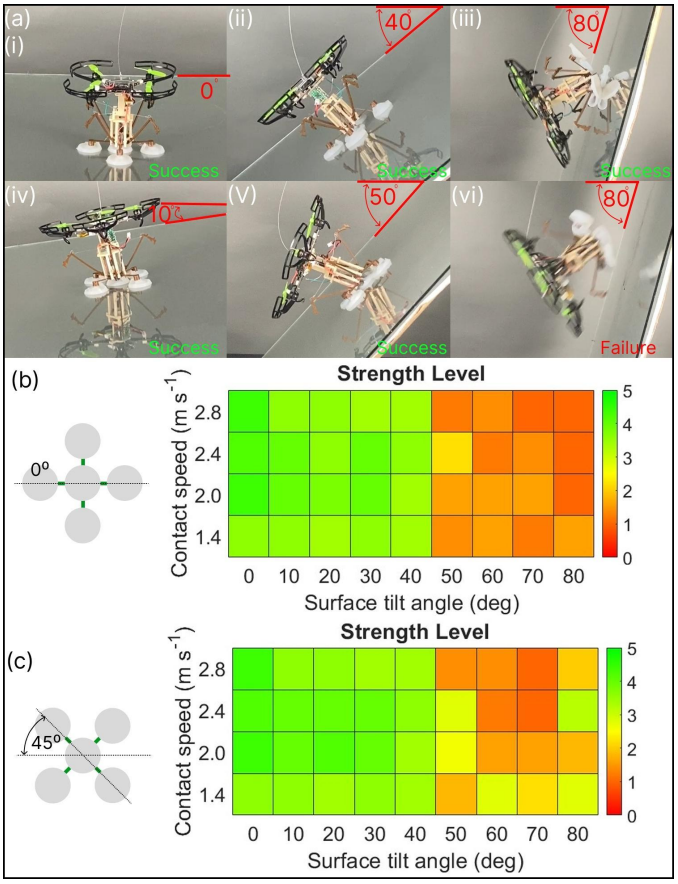


Fig. 12. Experimental results for the penta-cup mechanism. (a) Example perching at different tilt α and yaw angles β: (i) success at $\alpha = 0°$, $\beta = 0°$; (ii) success at $\alpha = 40°$, $\beta = 0°$; (iii) success at $\alpha = 80°$, $\beta = 0°$; (iv) success at $\alpha = 10°$, $\beta = 45°$; (v) success at $\alpha = 50°$, $\beta = 45°$; (vi) failure at $\alpha = 80°$, $\beta = 45°$. (b) Strength level variation for $\beta = 0°$. (c) Strength level variation for $\beta = 45°$.

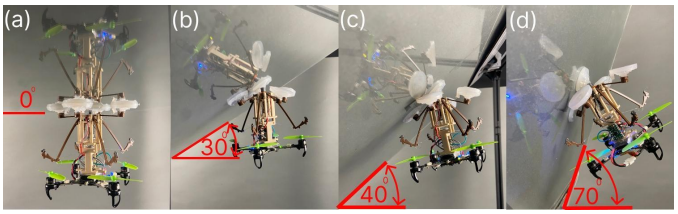


Fig. 13. Surface perching tests with drone flying for the penta-cup mechanism. (a) Success at $\alpha = 0°$, $\beta = 45°$. (b) Success at $\alpha = 30°$, $\beta = 45°$. (c) Success at $\alpha = 40°$, $\beta = 0°$. (d) Success at $\alpha = 70°$, $\beta = 0°$.

for three claw-end geometries: flat, triangle, and spike, each intended to enhance contact and grip. We measured holding force for cylindrical objects with four diameters (20, 25, 30, and 35 mm) for each design. The spike geometry consistently produces the highest force. The spike and triangle designs achieve 7.1 ± 1.2 N and 3.4 ± 0.6 N, respectively, for diameters up to 30 mm, followed by a significant reduction at 35 mm due to limited enclosure of larger cylinders. In contrast, the flat design exhibits a nearly constant low force of 0.8 ± 0.2 N across all diameters, attributed to limited friction. To assess performance on non-cylindrical perching objects, experiments are conducted using a cuboid geometry ($20 \times 20 \times 80$ mm).

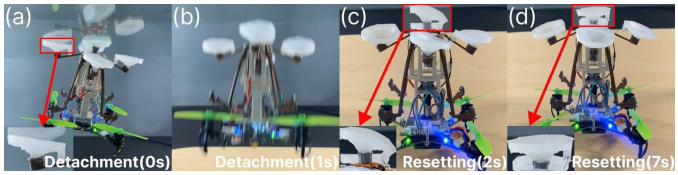


Fig. 14. Surface detachment. (a) Initial detachment of a suction cup (zoomed view shows the moment of detachment). (b) All cups detached, and the mechanism begins to fall. (c) Deformed cups immediately after detachment. (d) Suction cups restore to their original shape, ready to perch again.

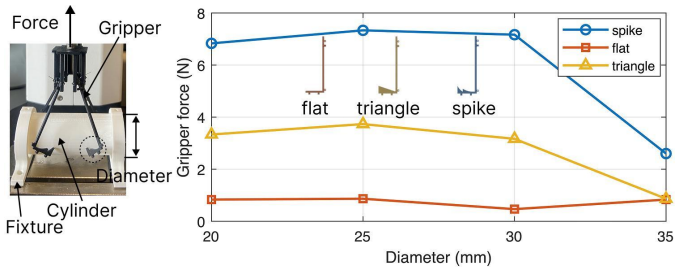


Fig. 15. Gripper force test for different cylinder diameters and three claw-end designs.

The spike design again achieves the highest average holding force (24.4 N), compared to 18.0 N for the triangle and 1.5 N for the flat design, enabling the mechanism to support loads of up to approximately 77 times its own weight.

We conduct flying experiments using the spike design to demonstrate branch perching on cylinders with diameters ranging from 15 to 40 mm. The mechanism successfully perches onto branch sections up to 35 mm, but fails at 40 mm, as successful perching requires claw enclosure of the cylinder (Fig. 16). At 40 mm, the mechanism cannot enclose the cylinder. For larger diameters, the claws' dimensions need to be adjusted to achieve successful perching.

D. Real world experiments

We finally demonstrate the hybrid mechanism in three real-world scenarios, including branch perching on a tree branch, and surface perching on a tilted whiteboard and the side-view mirror of a car. In all these experiments, the mechanism is mounted on top of the Crazyflie. When perching on a tree branch, the experiments highlight adaptability to wildlife environments: the mechanism's claws start open and, once positioned with the branch, close to enclose it (Fig. 17a). For perching to a car's side-view mirror, the mechanism approaches directly and attaches as long as it can be roughly controlled to contact the mirror (Fig. 17b). Finally, perching on a tilted whiteboard showcases applicability in indoor environments, where the mechanism attaches to the surface despite the large tilt angle (Fig. 17c). Overall, these experiments validate the mechanism's versatility across indoor and outdoor environments, suggesting the mechanism is well-suited for a range of real-world objects.

VI. CONCLUSIONS

This paper presents the design, analysis, and experimental evaluation of a perching mechanism that integrates branch

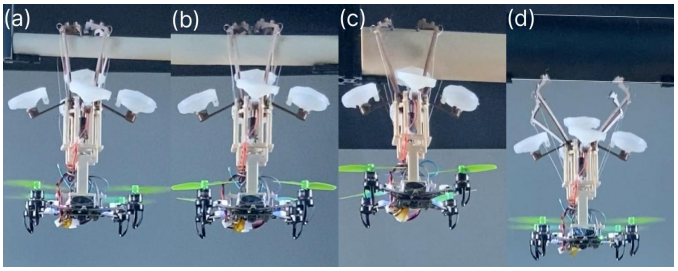


Fig. 16. Branch perching tests. (a) Successful perch at 15 mm. (b) Successful perch at 20 mm. (c) Successful perch at 35 mm. (d) Failed perch at 40 mm.

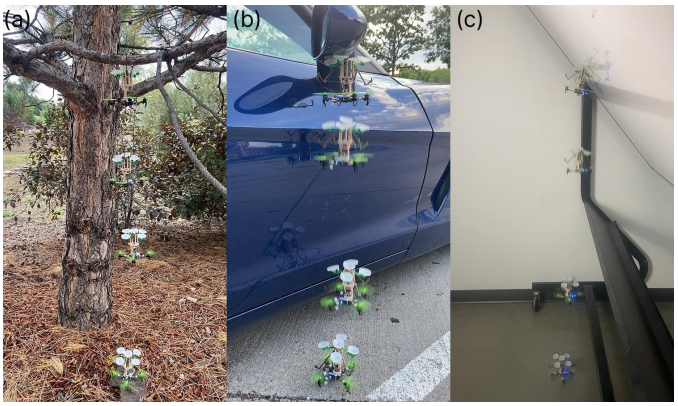


Fig. 17. Real-world experiments. (a) Tree branch perching. (b) Surface perching on a car side-view mirror. (c) Surface perching on a tilted whiteboard.

perching and surface perching. The key findings indicate improved handling of tilted surfaces and better perching capabilities: with the bistable design, it can successfully perch on flat surfaces with tilt angles up to 80° while maintaining adequate attachment strength. For branch perching, the mechanism can handle branch diameters up to 35 mm. We also show that the perching mechanism enables a miniature quadcopter to perch on real-world objects. Future work will focus on extending surface-perching capability to rough surfaces, broadening the range of compatible substrates [9].

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